Mathematical Methods for Finance

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Module 5 - Lesson 15

Variation, Riemann–Stieltjes integration and quadratic variation

Why Brownian paths force a new calculus

Learning outcomes for Lesson 15

By the end of the lesson students should be able to

1. compute total variation and quadratic variation in simple deterministic examples;
2. distinguish net change, first variation and quadratic variation;
3. recognise when a Riemann–Stieltjes integral is a classical pathwise integral;
4. explain why Brownian motion is not an ordinary bounded-variation integrator;
5. interpret semimartingales as the natural class for stochastic integration.

Intuition: Operating rule

Before computing, identify the object: pathwise variation, ordinary integral, quadratic variation, stochastic integral, Itô process or stochastic differential rule.

How today is organised

Block 1

14:30–15:20
Variation of paths, partitions, first examples and visual intuition.

Block 2

15:35–16:25
Riemann–Stieltjes integration, finite variation and semimartingales.

Block 3

16:40–17:30
Quadratic variation, Brownian intuition and exam-style exercises.

Exercise: During the lesson

Every exercise should end with one sentence explaining what the formula means.

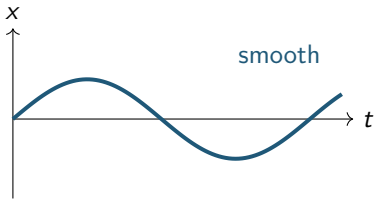
Block 1

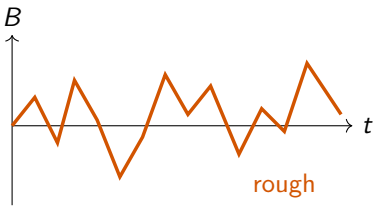
Variation of paths

The central question

Intuition: From smooth to rough

For a smooth curve, small increments are of order Δt. For Brownian motion, increments are of order $\sqrt{\Delta t}$.





Key novelty

The term $(\Delta B)^2$ is not negligible in the limit. It accumulates into dt.

Partitions and mesh

Definition: Partition

A partition of $[0, T]$ is a finite ordered set

$$\pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}.$$

Its mesh is

$$|\pi| = \max_i (t_i - t_{i-1}).$$

Exercise: Quick check $\star$

For $\pi = \{0, 0.1, 0.4, 0.7, 1\}$, compute $|\pi|$.

Worked example: Solution

The increments are $0.1, 0.3, 0.3, 0.3$. Hence $|\pi| = 0.3$.

Total variation

Definition: Variation along a partition

For a path $x : [0, T] \to \mathbb{R}$,

$$V_T(x; \pi) = \sum_{i=1}^{n} |x(t_i) - x(t_{i-1})|.$$

The total variation is

$$V_T(x) = \sup_{\pi} V_T(x; \pi).$$

Intuition: Meaning

Variation is the total distance travelled by the path, not only the distance between start and end.

Net change is not variation

Counterexample: False shortcut

It is generally false that

$$V_T(x) = |x(T) - x(0)|.$$

Example: Up and down path

If $x(0) = 0$, $x(1) = 1$, $x(2) = 0$, then the net change is 0, but the variation along the partition $\{0, 1, 2\}$ is

$$|1 - 0| + |0 - 1| = 2.$$

Worked example: monotone path

Worked example: Increasing path

If x is increasing on $[0, T]$, then

$$V_T(x) = x(T) - x(0).$$

Exercise: Basic computation $\star$

Compute the total variation of $x(t) = t^2$ on $[0, 3]$.

Worked example: Solution

The function is increasing on $[0, 3]$. Therefore

$$V_3(x) = x(3) - x(0) = 9.$$

Worked example: piecewise linear path

Exercise: Piecewise path ⋆⋆

A path is linear between the points

$$(0,1), \quad (1,3), \quad (2,-2), \quad (4,0).$$

Compute its total variation on $[0,4]$.

Worked example: Solution

For a piecewise linear path, sum the absolute vertical moves:

$$V_4(x) = |3-1| + |-2-3| + |0-(-2)| = 2+5+2 = 9.$$

Absolutely continuous paths

Definition: Absolutely continuous path

A path x is absolutely continuous if there is an integrable function $\dot{x}$ such that

$$x(t) = x(0) + \int_0^t \dot{x}(s)\, ds.$$

Theorem: Variation formula

If x is absolutely continuous, then

$$V_T(x) = \int_0^T |\dot{x}(s)|\, ds.$$

Exercise: derivative and variation

Exercise: ⋆⋆

Let $x(t) = t + \sin t$ on $[0, \pi]$. Show that x is increasing and compute $V_\pi(x)$.

Worked example: Solution

We compute

$$\dot{x}(t) = 1 + \cos t \geq 0 \quad \text{on } [0, \pi].$$

Hence x is increasing and

$$V_\pi(x) = x(\pi) - x(0) = \pi + \sin \pi - (0 + \sin 0) = \pi.$$

p-variation and quadratic sums

Definition: p-variation sums

For $p > 0$,

$$V_T^{(p)}(x;\pi) = \sum_i |x(t_i) - x(t_{i-1})|^p.$$

The case $p = 1$ is first variation; the case $p = 2$ is quadratic variation along the partition.

Intuition: Why $p = 2$ matters

For smooth paths, squared increments disappear. For Brownian paths, squared increments accumulate.

Quadratic sums: smooth path

Theorem: Smooth paths have zero quadratic variation

If x is continuously differentiable, then for partitions with $|\pi| \to 0$,

$$\sum_i \big(x(t_i) - x(t_{i-1})\big)^2 \longrightarrow 0.$$

Proof idea: Estimate

If $|x'| \le M$, then

$$|\Delta x_i| \le M \Delta t_i, \quad (\Delta x_i)^2 \le M^2 (\Delta t_i)^2.$$

Summing gives a bound of order $|\pi|$.

Exercise: quadratic sum of a linear path

Exercise: ⋆⋆

Let $x(t) = t$ on $[0, 1]$ and take the uniform partition with n intervals. Compute

$$\sum_{i=1}^{n} (\Delta x_i)^2.$$

Worked example: Solution

Each increment is $\Delta x_i = 1/n$. Hence

$$\sum_{i=1}^{n} (\Delta x_i)^2 = n \left(\frac{1}{n}\right)^2 = \frac{1}{n} \to 0.$$

Block 2

Riemann–Stieltjes integration and semimartingales

Riemann–Stieltjes sums

Definition: Riemann–Stieltjes integral

Given paths f and g, choose $\xi_i \in [t_{i-1}, t_i]$ and form

$$\sum_i f(\xi_i)\big(g(t_i) - g(t_{i-1})\big).$$

If the sums converge as $|\pi| \to 0$, the limit is written

$$\int_0^T f(t)\, dg(t).$$

Intuition: Integrator as a measuring device

The increment dg gives weight to the movement of g, not merely to elapsed time.

Ordinary integral as a special case

Example: $g(t) = t$

If $g(t) = t$, then $dg(t) = dt$ and

$$\int_0^T f(t)\, dg(t) = \int_0^T f(t)\, dt.$$

Exercise: Recognize the integral $\star$

Classify:

$$\int_0^1 t^2\, dt, \qquad \int_0^1 t\, d(t^2), \qquad \int_0^1 B_t\, dt.$$

Worked example: Answer

The first is ordinary. The second is Riemann–Stieltjes. The third is an ordinary time integral with random integrand.

Existence with bounded variation

Theorem: Sufficient condition

If f is continuous and g has bounded variation on $[0, T]$, then

$$\int_0^T f(t)\, dg(t)$$

exists as a pathwise Riemann–Stieltjes integral.

Interpretation

Classical integration works well when the integrator has finite accumulated movement.

Computation with differentiable integrator

Theorem: Reduction formula

If g is differentiable with integrable derivative, then

$$\int_0^T f(t)\,dg(t) = \int_0^T f(t)g'(t)\,dt.$$

Exercise: $\star\star$

Compute

$$\int_0^1 t\,d(t^2).$$

Worked example: Solution

Here $g(t) = t^2$, so $g'(t) = 2t$. Hence

$$\int_0^1 t\,d(t^2) = \int_0^1 2t^2\,dt = \frac{2}{3}.$$

Integration by parts

Theorem: Pathwise integration by parts

For sufficiently regular finite-variation paths,

$$f(T)g(T) - f(0)g(0) = \int_0^T f\, dg + \int_0^T g\, df.$$

Exercise: ⋆⋆

Use integration by parts to compute $\int_0^1 t^2\, dt$.

Worked example: Solution

Take $f(t) = t^2$, $g(t) = t$. Then $dg = dt$, so the integral is ordinary:

$$\int_0^1 t^2\, dt = \frac{1}{3}.$$

The formula checks: $1 = \frac{1}{3} + \int_0^1 t\, d(t^2) = \frac{1}{3} + \frac{2}{3}$.

Why Brownian motion breaks the classical picture

Counterexample: Brownian paths

Brownian paths are continuous, but almost surely have infinite total variation on every interval.

Intuition: Reason

A typical Brownian increment is of size $\sqrt{\Delta t}$. Over n intervals of length T/n, the sum of absolute moves is roughly

$$n\sqrt{T/n} = \sqrt{nT} \to \infty.$$

Consequence

We need another integral, built probabilistically rather than path by path through bounded variation.

Finite variation process

Definition: Finite variation process

A process A has finite variation on $[0, T]$ if, for almost every ω, the path $t \mapsto A_t(\omega)$ has finite total variation on $[0, T]$.

Example: Examples

- $A_t = \int_0^t a_s \, ds$ when $\int_0^T |a_s| \, ds < \infty$.
- A monotone counting path with finitely many jumps on $[0, T]$.
- A deterministic differentiable function of time.

Semimartingales

Definition: Semimartingale, informal version

A semimartingale can be decomposed as

$$X_t = X_0 + M_t + A_t,$$

where M is a local martingale and A is a finite-variation process.

Intuition: Why this class

Semimartingales are the natural integrators for stochastic calculus: they contain a predictable drift-like part and an irregular martingale part.

Examples of semimartingales

Example: Basic examples

- Brownian motion: $B_t = 0 + B_t + 0$.
- Deterministic finite-variation path: $A_t = 0 + 0 + A_t$.
- Itô process: $X_t = X_0 + \int_0^t b_s \, ds + \int_0^t \sigma_s \, dB_s$.
- Poisson process: $N_t = (N_t - \lambda t) + \lambda t$.

Exercise: $\star\star$

Identify the martingale part and finite-variation part of $X_t = 3 + t + 2B_t$.

Worked semimartingale decomposition

Worked example: Solution

Write

$$X_t = 3 + 2B_t + t.$$

Thus $X_0 = 3$, the martingale part is

$$M_t = 2B_t,$$

and the finite-variation part is

$$A_t = t.$$

Intuition: Exam habit

Always separate initial value, martingale noise and accumulated drift.

Doob decomposition: discrete intuition

Theorem: Discrete-time idea

A suitable integrable adapted process can often be decomposed into

$$X_n = X_0 + M_n + A_n,$$

where M is a martingale and A is predictable finite variation.

Intuition: Meaning

Every movement is split into an unpredictable martingale innovation and a predictable accumulated part.

Exercise: predictable plus martingale

Exercise: ⋆⋆⋆

Let $X_n = n + S_n$, where S_n is a symmetric random walk. Give a decomposition into finite-variation predictable part and martingale part.

Worked example: Solution

The random walk S_n is a martingale. The deterministic process $A_n = n$ is predictable and finite variation. Hence

$$X_n = X_0 + M_n + A_n = 0 + S_n + n.$$

Block 3

Quadratic variation and exam training

Quadratic variation

Definition: Quadratic variation, informal definition

For a process X, the quadratic variation is the limit, when it exists, of sums

$$\sum_i \left(X_{t_i} - X_{t_{i-1}}\right)^2.$$

It is denoted by $\langle X \rangle_t$.

Intuition: What it measures

Quadratic variation measures accumulated second-order movement.

Brownian quadratic variation

Theorem: Main fact

For Brownian motion,

$$\sum_i (B_{t_i} - B_{t_{i-1}})^2 \longrightarrow T$$

in probability along partitions of $[0, T]$ whose mesh goes to zero. Thus

$$\langle B \rangle_T = T.$$

Intuition: Heuristic

Each increment has variance Δt_i, so each square is typically of size Δt_i. Summing gives T.

Exercise: Brownian scaling heuristic

Exercise: ⋆⋆

On $[0, T]$, divide time into n equal intervals. If ΔB_i has variance T/n, compute

$$\mathbb{E}\left[\sum_{i=1}^{n}(\Delta B_i)^2\right].$$

Worked example: Solution

Since $\mathbb{E}[(\Delta B_i)^2] = T/n$,

$$\mathbb{E}\left[\sum_{i=1}^{n}(\Delta B_i)^2\right] = \sum_{i=1}^{n}\frac{T}{n} = T.$$

Covariation

Definition: Quadratic covariation

For two processes X, Y,

$$\langle X, Y \rangle_t$$

is obtained as the limit of

$$\sum_i \left(X_{t_i} - X_{t_{i-1}}\right)\left(Y_{t_i} - Y_{t_{i-1}}\right).$$

Example: Brownian multiples

If $X_t = aB_t$ and $Y_t = bB_t$, then

$$\langle X, Y \rangle_t = ab\, t.$$

Finite variation and quadratic variation

Theorem: Continuous finite-variation paths

If A is continuous and has finite variation, then

$$\langle A \rangle_t = 0.$$

Example: Drift part

For $A_t = \int_0^t a_s \, ds$, we have

$$\langle A \rangle_t = 0.$$

Memory aid

Drift contributes to first-order movement. Brownian martingale parts contribute to second-order movement.

Quadratic variation of an Itô-like process

Example: Simple process

Let

$$X_t = x_0 + bt + \sigma B_t.$$

Then

$$\langle X \rangle_t = \sigma^2 t.$$

Exercise: $\star\star$

Find $\langle X \rangle_t$ for $X_t = 5 + 3t - 2B_t$.

Worked example: Solution

Only the Brownian coefficient matters:

$$\langle X \rangle_t = (-2)^2 t = 4t.$$

Exam-style comparison table

Object	First variation	Quadratic variation
Smooth path	finite	zero
Continuous finite variation	finite	zero
Brownian motion	infinite	t
$bt + \sigma B_t$	infinite if $\sigma \neq 0$	$\sigma^2 t$
Pure jump with finite jumps	finite	sum of squared jumps

Exercise: Quick oral check

Why can a process have infinite first variation but finite quadratic variation?

Exercise: classify objects

Exercise: ⋆⋆

Classify each expression:

$$\int_0^T f(t)\,dt, \quad \int_0^T f(t)\,dA_t, \quad \langle B \rangle_T, \quad \int_0^T H_t\,dB_t.$$

Which ones are already classical and which require stochastic calculus?

Worked example: Solution

The first is ordinary. The second is pathwise Riemann–Stieltjes if A has bounded variation. $\langle B \rangle_T$ is a stochastic path property. The last is an Itô stochastic integral and requires the construction of Lesson 16.

Exercise: total versus quadratic variation

Exercise: ⋆⋆

A path observed at times $0, 1, 2, 3, 4$ has values

$$0, \quad 2, \quad 1, \quad 4, \quad 3.$$

Compute the first-variation sum and quadratic-variation sum on this partition.

Worked example: Solution

The increments are $2, -1, 3, -1$. Hence

$$\sum |\Delta x_i| = 2 + 1 + 3 + 1 = 7,$$

and

$$\sum (\Delta x_i)^2 = 4 + 1 + 9 + 1 = 15.$$

Exercise: continuous drift plus Brownian noise

Exercise: $\star\star\star$

Let

$$X_t = \int_0^t (1+s)\,ds + 3B_t.$$

Identify the finite-variation part and compute $\langle X \rangle_t$.

Worked example: Solution

The finite-variation part is

$$A_t = \int_0^t (1+s)\,ds = t + \frac{1}{2}t^2.$$

The martingale part is $3B_t$. Therefore

$$\langle X \rangle_t = \langle 3B \rangle_t = 9t.$$

Common mistakes

Counterexample: Mistakes to avoid

- Confusing $|x(T) - x(0)|$ with total variation.
- Treating $dB_t/\,dt$ as an ordinary derivative.
- Forgetting that Brownian first variation is infinite.
- Forgetting that continuous finite-variation terms have zero quadratic variation.
- Computing $\langle bt + \sigma B \rangle_t$ as $(b + \sigma)^2 t$ instead of $\sigma^2 t$.

End-of-lesson checklist

Can you do these?

1. Compute total variation for monotone and piecewise linear paths.
2. Recognize when a Riemann–Stieltjes integral is classical.
3. Explain why Brownian paths need stochastic integration.
4. Compute $\langle bt + \sigma B \rangle_t$.
5. Separate finite-variation and martingale parts in simple examples.

Exercise: Suggested practice

From the Module 5 exercise bank: variation, Riemann–Stieltjes, semimartingale and quadratic-variation exercises.